

Product Carbon Footprint Study Report

Product Name: synthetic rubber

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: synthetic rubber

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg synthetic rubber

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg synthetic rubber.

2. System Boundary

synthetic rubber production in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of synthetic rubber produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **3.120 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

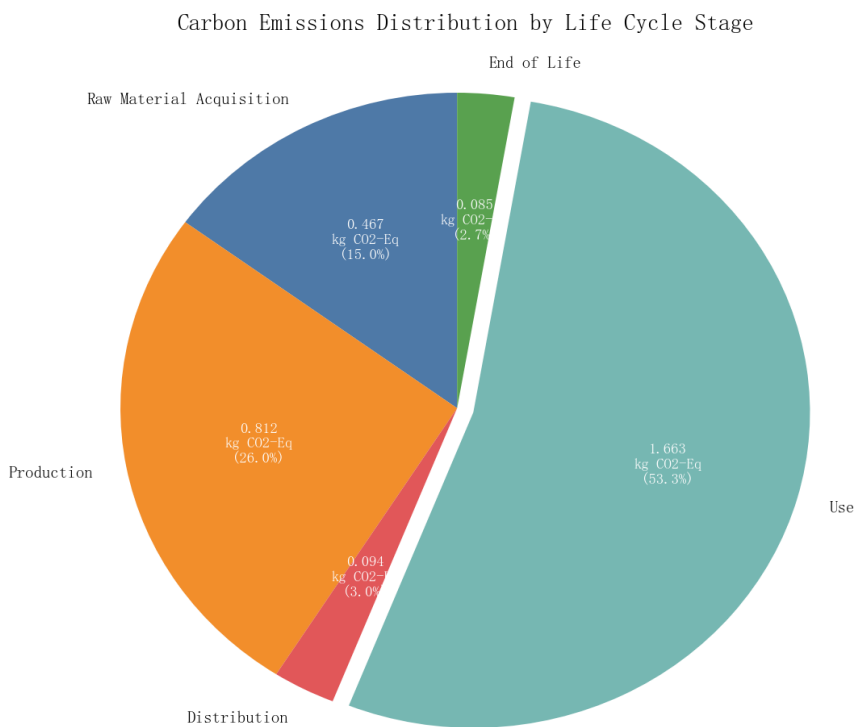


Table 1: Carbon Emissions by Life Cycle Stage

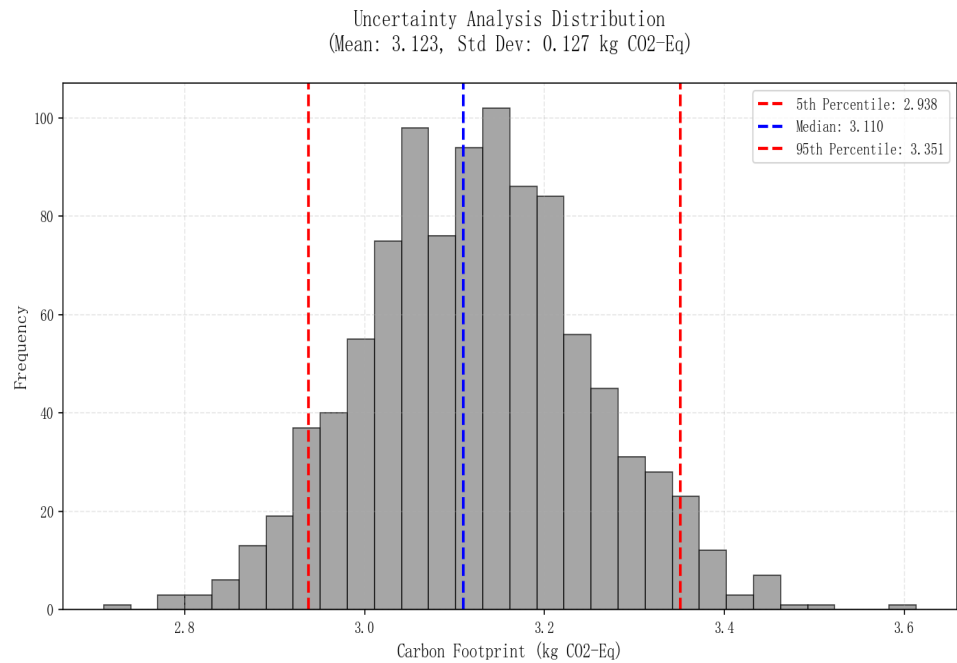
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.467	14.96
Production	0.812	26.01
Distribution	0.094	3.00

Use	1.663	53.29
End of Life	0.085	2.74
Total	3.120	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	3.123
Standard Deviation	0.127
Coefficient of Variation (CV)	4.08%
Median	3.110
5th Percentile	2.938
95th Percentile	3.351

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the synthetic rubber product stands at 3.120 kg CO₂-Eq, representing a moderate emission intensity typical for petrochemical-derived materials in industrial applications. This aggregated value encompasses all lifecycle stages and provides a benchmark for comparing environmental performance against alternative materials or production scenarios.

Lifecycle stage contributions reveal a clear dominance by the use phase at 1.663 kg CO₂-Eq (53.3%), followed by production at 0.812 kg CO₂-Eq (26.0%) and raw materials at 0.467 kg CO₂-Eq (15.0%). Distribution and end-of-life stages account for only 3.0% and 2.7%, respectively, indicating that emissions are heavily concentrated in operational and manufacturing activities rather than logistics or disposal.

Major emission sources are therefore the use phase, driven by product application demands, and the production stage, linked to energy-intensive processing. Targeted interventions in these areas would address over 79% of the total footprint.

2. Statement of Assumptions and Limitations

This study relies on several key assumptions, including the use of average European energy mixes for production processes, standard transportation distances for distribution, and typical product lifetimes for the use phase. It also assumes that end-of-life treatment follows conventional waste management practices without advanced recycling credits.

Limitations arise from reliance on secondary data sources such as generic industry databases, which may not fully capture site-specific variations in raw material sourcing or manufacturing technologies. Data quality is further constrained by temporal mismatches and potential aggregation of emission factors that overlook regional differences in feedstock composition.

System boundary restrictions encompass a cradle-to-grave scope limited to the defined stages, excluding indirect upstream impacts from capital equipment, employee commuting, and potential rebound effects during product use. Certain co-product allocations and biogenic carbon flows have been omitted to maintain consistency with the functional unit.

3. Suggestions for Improvement

- Implement energy efficiency upgrades and transition to renewable electricity sources in the production stage to directly cut the 26% contribution from manufacturing.
- Enhance product durability and application-specific design (e.g., lower rolling resistance formulations) to reduce emissions during the dominant use phase.
- Source lower-carbon raw materials or increase recycled feedstock content to lower the 15% impact from raw material extraction.
- Conduct primary data collection at production facilities and supplier sites to improve accuracy beyond generic database values.

- Perform sensitivity analyses on key parameters such as energy mix and product lifetime to refine reduction pathway prioritization.

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